DAY 04 | POST-READ

Architecture and MongoDB: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Matplotlib	plot, bar, hist, scatter, pie. Add title, labels, legend. show() or savefig().
Client server	Clients send requests; servers process and respond, usually with JSON over HTTP.
API	A contract describing what you can ask for and what you get back.
Layers	Routes, services, repositories. Each has one job and can be tested alone.
NoSQL	Flexible documents instead of fixed tables. Good for nested, evolving data.
BSON	Binary JSON used by MongoDB. Adds types such as ObjectId and dates.
Create	insert_one, insert_many
Read	find_one, find with filters ($gt, $lt, $in, $regex) and projections
Update	update_one, update_many with $set, $inc, $push
Delete	delete_one, delete_many. Always test the filter with find first.

Part B: Going beyond today

1. Subplots: several charts in one figure

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))
ax1.bar(cats, totals)
ax2.plot(months, sales)
fig.tight_layout()
```

2. Embedding vs referencing in MongoDB

Embed related data inside a document when it is read together and does not grow without limit (an order with its line items). **Reference** by _id when data is shared or very large (an order pointing to a customer).

3. Upserts

```
products.update_one({"sku": "P-100"}, {"$set": {"price": 499}}, upsert=True)
```

If no document matches, MongoDB inserts one. Useful for syncing data.

4. Sorting and limiting

```
for p in products.find().sort("price", -1).limit(3):
    print(p["name"])
```

5. Connection strings and secrets

A MongoDB Atlas connection string contains a password. Never hard-code it or push it to GitHub. Read it from an environment variable: os.environ["MONGO_URI"]. We will cover secrets properly on Day 33.

6. Stateless servers

Good APIs do not remember anything between requests; every request carries what it needs. This lets you run many copies of a server behind a load balancer.

Part C: Check yourself

- What is the role of an API between a client and a database?
- Name two differences between SQL and document databases.
- What does BSON add that JSON does not have?
- What is the difference between $set and $inc?
- Why should you test a filter with find before delete_many?

Part D: Practice ideas for tonight

- Create a free MongoDB Atlas cluster and connect to it from Colab using the Option B cell.
- Store your weekly expenses as documents and chart spending per category.
- Design documents for a movie app: movies, reviews and users. Decide what to embed and what to reference.

Coming up on Day 5: REST principles, HTTP methods and status codes, and your first FastAPI app. There is also a 2 hour assessment on Week 1.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today